Jonghun Kim

Email: iproj222@gmail.com **Github**: github.com/jongdory **Homepage**: Jongdory.github.io

Research Interests

Medical Image Analysis, Medical Image Translation, Generative Models, Radiomics

Education

Hanyang University March 2015 — February 2022
B.S. in Computer Science & Engineering
Sungkyunkwan University March 2022 — February 2024
M.S. in Electrical and Computer Engineering
Sungkyunkwan University March 2024 — Current
Ph.D. in Electrical and Computer Engineering

Journal Publications

* First authors, † Corresponding authors. If absent, first/last authors are first/corresponding.

[**J7**] Dual-conditioned diffusion model with anatomical guidance for geometric distortion correction in prostate MRI
Inye Na, Qi Miao, **Jonghun Kim**, Kyunghyun Sung†, Hyunjin Park†
European Radiology Experimental, 2026

[**J6**] LS-MAT: Lifespan structural magnetic resonance imaging synthesis for microstructural covariance profile analysis toolbox
Hokyun Kim, **Jonghun Kim**, Mansu Kim, Hyunjin Park, Bo-yong Park
NeuroImage, 2026

[**J5**] Visual prompt tuning for task-flexible medical image synthesis
Jonghun Kim, Hyunjin Park
Computer Methods and Programs in Biomedicine, 2026

[**J4**] Enhancing intracranial vessel segmentation using diffusion models without manual annotation for 3D Time-of-Flight Magnetic Resonance Angiography
Jonghun Kim, Inye Na, Jiwon Chung, Ha-Na Song, Kyungseo Kim, Seongvin Ju, Mi-Yeon Eun, Woo-Keun Seo†, Hyunjin Park†
Computerized Medical Imaging and Graphics, 2025

[**J3**] Visual prompt tuning for task-flexible medical image synthesis
Jonghun Kim, Inye Na, Junmo Kwon, Woo-Keun Seo, Hyunjin Park
Computer Methods and Programs in Biomedicine, 2025

[**J2**] Enhancing radiomics features via a large language model for classifying benign and malignant breast tumors in mammography
Sinyoung Ra, **Jonghun Kim**, Inye Na, Eun Sook Ko, Hyunjin Park
Computer Methods and Programs in Biomedicine, 2025

[**J1**] Leveraging segmentation-guided spatial feature embedding for overall survival prediction in glioblastoma with multimodal magnetic resonance imaging
Junmo Kwon, **Jonghun Kim**, Hyunjin Park
Computer Methods and Programs in Biomedicine, 2024

Conference Publications

* First authors, † Corresponding authors. If absent, first/last authors are first/corresponding.

[**C19**] Mitigating Heterogeneity with Spatial Routing for 3D Medical Image Translation
Jonghun Kim, Hyunjin Park
IEEE BHI 2026

[**C18**] Anatomical Prior-Guided Routing for 3D Lung Tumor Segmentation
Uihyeon Maeng*, **Jonghun Kim***, Hyunjin Park
ACCV 2026

[**C17**] Mixture of Multi-Sequence Feature Difference for Contrast-Enhanced MRI Synthesis
Jonghun Kim Junmo Kwon, Hyunjin Park
ACCV 2026

[**C16**] Posterior Samplings are Missing Modalities Generators for Medical Image Translation
Jonghun Kim
ECCV 2026

[**C15**] Estimated Age-Guided Cerebral Microbleed Segmentation
Junmo Kwon, **Jonghun Kim**, Taehyeon Kim, Hyunjin Park
MICCAI 2026

[**C14**] Visual Instruction-Finetuned Language Model for Versatile Brain MR Image Tasks
Jonghun Kim, Sinyoung Ra, Hyunjin Park
ICPR 2026

[**C13**] Multi-LLM Collaborative MRI Report Generation for Visual Instruction Tuning in Brain Oncology
Sinyoung Ra*, **Jonghun Kim***, Hyunjin Park
IEEE EMBC 2026

[**C12**] Harmonizing Multi-Domain Heterogeneity in Medical Imaging via Gaussian Mixture Model
Gyeongdeok Jo, **Jonghun Kim**, Nejung Rue, Hyunjin Park
IEEE BIBM 2025

[**C11**] Privacy Preserving Chest X-Ray Classification in Latent Space with Homomorphically Encrypted Neural Inference
Jonghun Kim, Gyeongdeok Jo, Sinyoung Ra, Hyunjin Park
MICCAI 2025

[**C10**] PMIL: Prompt Enhanced Multimodal Integrative Analysis of fMRI Combining Functional Connectivity and Temporal Latency
Hyoungshin Choi, **Jonghun Kim**, Jiwon Chung, Bo-yong Park, Hyunjin Park
MICCAI 2025

[**C9**] Blood Pressure Assisted Cerebral Microbleed Segmentation via Meta-matching
Junmo Kwon, **Jonghun Kim**, Taehyeon Kim, Sang Won Seo, Hwan-ho Cho, Hyunjin Park
MICCAI 2025

[**C8**] Multimodal structural MRI synthesis pipeline across age
Ho-Kyun Kim, **Jonghun Kim**, Hyunjin Park, Bo-Yong Park
IEEE EMBC 2025

[**C7**] Simulating Post-Neoadjuvant Chemotherapy Breast Cancer MRI via Diffusion Model with Prompt Tuning
Jonghun Kim, Hyunjin Park
IEEE ISBI 2025

[**C6**] Tumor Synthesis Conditioned on Radiomics
Jonghun Kim, Inye Na, Eun Sook Ko, Hyunjin Park
WACV 2025

[**C5**] DeAFusion: Detail-Aware Image Fusion for Whitematter Hyperintensity Segmentation
Gyeongdeok Jo, **Jonghun Kim**, Bo-Yong Park, Hyunjin Park
IEEE BIBM 2024

[**C4**] Radiomicsfill-mammo: Synthetic mammogram mass manipulation with radiomics features
Inye Na, **Jonghun Kim**, Eun Sook Ko, Hyunjin Park
MICCAI 2024

[**C3**] Domain Aware Multi-Task Pre-Training of 3D Swin Transformer for Brain MRI
Jonghun Kim*, Mansu Kim*, Hyunjin Park
ACCV 2024

[**C2**] Radiomics-guided Multimodal Self-attention Network for Predicting Pathological Complete Response in Breast MRI
Jonghun Kim, Hyunjin Park
IEEE ISBI 2024

[**C1**] Adaptive latent diffusion model for 3d medical image to image translation: Multi-modal magnetic resonance imaging study
Jonghun Kim, Hyunjin Park
WACV 2024

Award

First Place Award in COSMOS 2022 Grand Challenge
Jonghun Kim, Inye Na, Junmo Kwon, Hyunjin Park
SMRA 2022 and MICCAI 2022